

Dry Run Exercises

Unit 2 Variables, Assignment and Expressions

Complete the dry run table for each fragment of code given in the questions below. **Give** the value(s) printed to the screen when each is executed.

Follow the module dry run style guide in your answers.

1.

```
int x = 22;           // L1
int y = 16;           // L2
x = y;                // L3
System.out.println(x + "." + y); // L4
```

Complete the dry run table (you may not need all rows):

LINE	x	y	printed

Write EXACTLY what is printed here

2.

```
int num1 = 1;           // L1
int num2 = 3;           // L2
num1 = num2 + 1;        // L3
System.out.println(num1 + "." + num2); // L4
```

Complete the dry run table (you may not need all rows):

LINE	num1	num2	printed

Write EXACTLY what is printed here

3.

```
int a = 2;              // L1
int b = 4;              // L2
b = a + a + a;          // L3
System.out.println(a);  // L4
System.out.println(b);  // L5
```

Complete the dry run table (you may not need all rows):

LINE	a	b	printed

Write EXACTLY what is printed here

4.

```
int a = 4;           // L1
int b = 2;           // L2
b = a * a;           // L3
System.out.print(a); // L4
System.out.print(b); // L5
```

Complete the dry run table (you may not need all rows):

LINE	a	b	printed

Write EXACTLY what is printed here

5.

```
String colour1 = "BLUE";           // L1
String colour2 = "RED";             // L2

System.out.println(colour2 + colour1); // L3
```

Complete the dry run table (you may not need all rows):

LINE	colour1	colour2	printed

Write EXACTLY what is printed here

6.

```
String colour1 = "BLUE";           // L1
String colour2 = "RED";             // L2
String temp = "";                   // L3

temp = colour1;                     // L4
colour1 = colour2;                  // L5
colour2 = temp;                     // L6

System.out.println(colour1);        // L7
System.out.println(colour2);        // L8
```

Complete the dry run table (you may not need all rows):

LINE	colour1	colour2	temp	printed

Write EXACTLY what is printed here

--

7.

```
int y = 7;          // L1
int x = 8;          // L2
```

```
x = y;              // L3
y = x;              // L4
```

```
System.out.println(x);    // L5
System.out.println(y);    // L6
```

Complete the dry run table (you may not need all rows):

LINE	y	x	printed

Write EXACTLY what is printed here

8.

```
int x = 1;           // L1
```

```
x = 4;              // L2
```

```
x = 6;              // L3
```

```
System.out.println(x);      // L4
```

Complete the dry run table (you may not need all rows):

LINE	x	printed

Write EXACTLY what is printed here

--

9.

```
int x = 7;          // L1
int y = 8;          // L2
int z = 9;          // L3
```

```
x = y;              // L4
y = z;              // L5
z = x;              // L6
```

```
System.out.println(x + y + z);    // L7
```

Complete the dry run table (you may not need all rows):

LINE	x	y	z	printed

Write EXACTLY what is printed here

10.

```
char a = 'a';           // L1
char b = 'b';           // L2
char c = 'c';           // L3

a = b;                  // L4
System.out.println(a + "," + c);    // L5

b = c;                  // L6
c = a;                  // L7

System.out.println(b + ", " + c);    // L8
```

Complete the dry run table (you may not need all rows):

LINE	a	b	c	printed

Write EXACTLY what is printed here

11.

```
String s1 = "rats";           // L1
String s2 = "and";           // L2
String s3 = "cats";          // L3
s1 = s1 + s2;                // L4
s2 = s2 + s3;                // L5
s2 = s3;                     // L6
System.out.println(s3);      // L7
```

Complete the dry run table (you may not need all rows):

LINE	s1	s2	s3	printed

Write EXACTLY what is printed here

12.

```
String s1 = "cats";           // L1
String s2 = "eat";           // L2
String s3 = "mice";          // L3
s1 = s1 + s2;                 // L4
s2 = s2 + s3;                 // L5
s3 = s2;                      // L6
System.out.println(s3);       // L7
```

Complete the dry run table (you may not need all rows):

LINE	s1	s2	s3	printed

Write EXACTLY what is printed here